

Cross-Modal Knowledge Distillation for Chest Radiographic Diagnosis via Embedding Expansion, Reconstruction, and Classification

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Abstract. Deep learning models now can achieve expert-level performance for chest radiographic (CXR) diagnosis, but they require pre-training with large-scale image datasets and categorical labels. Such labels are, however, frequently incomplete and commonly fail to capture the full spectrum of clinical knowledge embedded in radiology reports issued by expert radiologists. Large language models (LLMs) can encode text into machine-understandable semantic representations, providing an opportunity for cross-modal knowledge distillation, thereby enhancing vision-based CXR diagnosis models. We propose **CREED**—**C**lassifying and **R**econstructing **E**xpanded **E**mbeddings for cross-modal knowledge **D**istillation to address this shortcoming. CREED has three learning objectives: (1) *embedding reconstruction* to preserve fine-grained language information, (2) *diagnostic classification* to bridge the modality gap, and (3) *KL-divergence minimization* to enforce alignment between vision and language embeddings. Extensive experiments across diverse diagnostic tasks and on report generation demonstrate that CREED consistently outperforms baseline and state-of-the-art (SOTA) fully-/self-supervised models. The results highlight the effectiveness of cross-modal knowledge distillation from LLMs for enhancing CXR diagnosis by infusing clinically rich semantics into vision models, ultimately improving diagnostic accuracy, interpretability, and generalization across diverse medical imaging tasks. The codes and pretrained models are available at [GitHub.com/JLiangLab/CREED](https://github.com/JLiangLab/CREED).

Keywords: Cross-Modal Knowledge Distillation · Chest Radiography.

1 Introduction

Automated chest radiographic (CXR) analysis is rapidly becoming a crucial tool in medical imaging, leveraging deep learning models to assist radiologists

in detecting and classifying thoracic diseases [10,1,14]. While these models have demonstrated expert-level performance, they predominantly rely on categorical labels, which are often incomplete and fail to capture the full spectrum of clinical knowledge embedded in reports issued by expert radiologists [22,15,5,17,19,18] (see Fig. 1). As a result, these models may lack the nuanced understanding required for appropriate medical decision-making [29]. The advancement of LLMs presents a promising opportunity to bridge this gap, because LLMs can effectively process and summarize expert knowledge within radiology reports, encoding them into machine-understandable semantic representations (*i.e.*, language embeddings, which are information-rich numerical vectors capturing clinical semantics). This capability enables the potential for cross-modal knowledge distillation, transferring detailed clinical information within reports via LLMs to vision-based models. However, a significant challenge arises due to the substantial domain gap between textual language embeddings contained within reports and visual feature embeddings from images, making direct alignment difficult. To address this critical limitation, this work explores a critical question: *How can we effectively distill detailed clinical semantic knowledge embedded in radiology reports via LLMs into vision-based CXR diagnosis models?*

We propose **CREED**, a framework that **C**lassifies and **R**econstructs **E**xpanded **E**mbeddings for cross-modal knowledge **D**istillation, as illustrated in Fig. 2, enabling the transfer of clinically relevant knowledge with detailed semantics from LLMs into vision-based CXR diagnosis models. CREED distills rich clinical semantics from LLM-generated embeddings into a vision model through three learning objectives: (1) *embedding reconstruction* to preserve fine-grained language information, (2) *diagnostic classification* to bridge the modality gap, and (3) *KL-divergence minimization* to enforce alignment between vision and language embeddings. We evaluated CREED across a variety of diagnostic tasks, including long-tailed, multi-class, multi-label, and binary classification (Table 1), and demonstrated that CREED consistently outperforms or matches state-of-the-art (SOTA) supervised and self-supervised models. Our results highlight CREED’s effectiveness for enhancing CXR diagnosis through cross-modal knowledge transfer. Additionally, CREED improves performance in radiology report generation (Table 2), showcasing its ability to learn detailed clinical semantic knowledge from LLMs.

Unlike existing visual-language pretraining models that align image and text representations using paired medical data (e.g., CXRs and radiology reports) through contrastive learning objectives [23,28,3], or the approach that incorporate medical knowledge graphs to encode structured pathology relationships [30], CREED distills knowledge from embeddings generated by frozen LLMs based on radiology reports, utilizing embedding reconstruction alongside a classification task with diagnostic labels to bridge the domain gap effectively.

Through this work, we have made the following contributions: (1) A model that distills clinical knowledge generated by LLMs based on radiology reports; (2) A framework, comprising an encoder, decoder, and classifier, shared between the LLM and the vision model, which expands, reconstructs, and classifies the

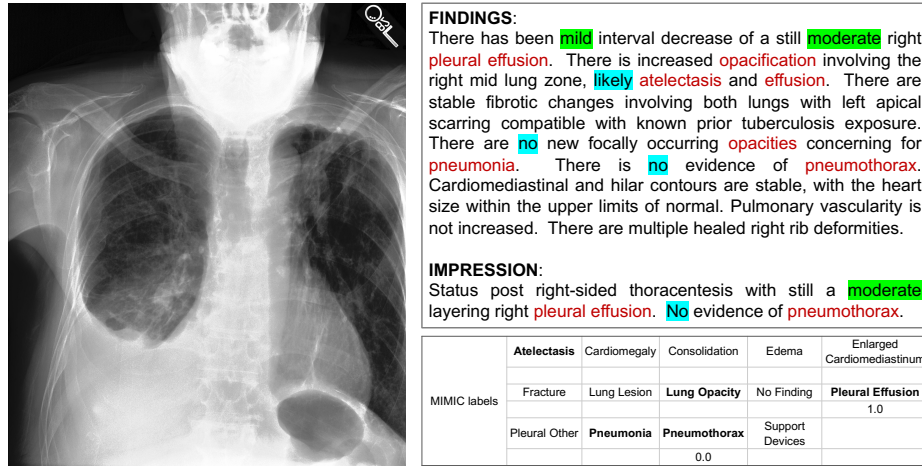


Fig. 1. Conventional training for chest radiographic diagnosis models primarily relies on structured labels (right bottom table), which are often incomplete, overlooking the rich clinical information embedded in reports created by expert radiologists. Radiology reports provide detailed descriptions of abnormalities, their severity, and contextual findings (highlighted with colors) that are not explicitly captured by categorical labels. Our approach distills essential knowledge contained within radiology reports into the vision model, enhancing its ability to comprehend nuanced diagnostic patterns beyond predefined labels, ultimately improving diagnostic performance.

embeddings to align the language and vision domains; (3) Comprehensive experiments that evaluate CREED on various diagnostic tasks and report generation, demonstrating its enhanced capability for CXR diagnosis.

2 Method

Architectural Components. The CREED framework is designed to distill knowledge from an LLM into a vision model by leveraging CXR images and their corresponding reports. At its core, CREED introduces a shared framework that operates between the LLM and the vision model, consisting of three key components: encoder, decoder, and classifier. The encoder expands input embeddings into a higher-dimensional latent space, in contrast to conventional autoencoders that compresses inputs. This expansion aligns embeddings from vision and language domains into a shared latent space, effectively integrating textual and visual semantics to enable cross-modal knowledge distillation. The intuition behind this design is to construct representations that fuse vision–language mutual information within a richer, higher-dimensional space. The decoder reconstructs the input language embeddings from the expanded latent space with either language or vision embeddings, ensuring knowledge transfer from the LLM to the vision model. The classifier maps vision and language embeddings into a shared

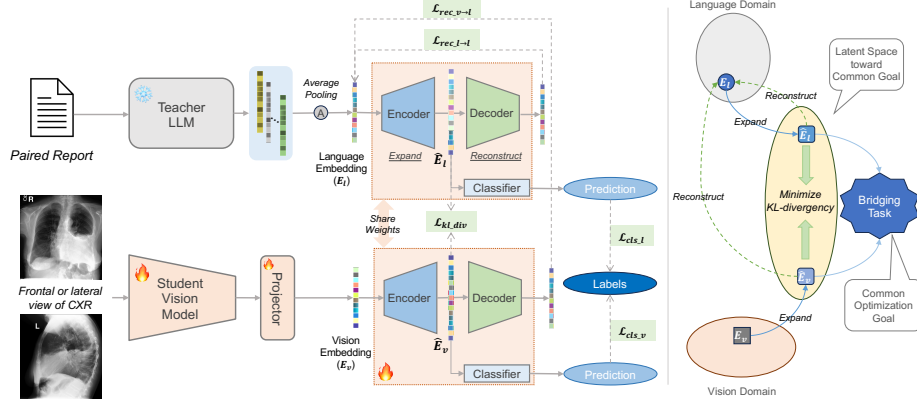


Fig. 2. CREED framework distills knowledge embedded in radiology reports via a frozen teacher LLM into a student vision model. Designed as a framework *shared* between the LLM and the vision model, CREED consists of an encoder, a decoder, and a linear classifier. Unlike conventional autoencoders that compress input into a lower-dimensional latent space, the CREED encoder expands input embeddings into a higher dimension. When processing vision embeddings, this expansion should integrate knowledge from the language domain, enhancing representation learning. To achieve this, we introduce three learning objectives that bridge vision and language embeddings: a reconstruction task, a classification task, and KL-divergence minimization. The reconstruction task employs a decoder to recover the original language embedding from the expanded language or vision embeddings ($\mathcal{L}_{rec_l \rightarrow l} + \mathcal{L}_{rec_v \rightarrow l}$). By requiring the expanded vision embedding to reconstruct the language embedding, CREED is forced to distill essential knowledge from the LLM. To facilitate cross-modal distillation between heterogeneous models and data, we introduce a bridging classification task as a common optimization goal across both modalities. By using diagnostic labels ($\mathcal{L}_{cls_l} + \mathcal{L}_{cls_v}$), this task aligns input embeddings from different modalities into a shared latent space. Additionally, we further enforce similarity between the latent spaces by minimizing the KL-divergence between the two expanded embeddings ($\mathcal{L}_{kl-div}$). The conceptual framework of CREED is illustrated on the right side.

diagnostic label space, unifying heterogeneous modalities through a common optimization objective.

Learning Objectives. To optimize cross-modal distillation, we introduce three learning objectives that bridge vision and language embeddings:

Embedding Reconstruction Task. The decoder reconstructs language embeddings from both modalities, facilitating knowledge distillation from the LLM to the vision model:

$$\mathcal{L}_{rec} = \mathcal{L}_{rec_l \rightarrow l}(E_l, Decoder(\hat{E}_l)) + \mathcal{L}_{rec_v \rightarrow l}(E_l, Decoder(\hat{E}_v)) \quad (1)$$

where $\mathcal{L}_{rec_l \rightarrow l}$ and $\mathcal{L}_{rec_v \rightarrow l}$ correspond to language-to-language and vision-to-language embedding reconstruction losses, respectively. Here, E_l represents

the original language embedding from the LLM, while $\hat{E}_l$ and $\hat{E}_v$ denote the expanded language and vision embeddings, respectively.

Bridging Classification Task. A shared classifier aligns diagnostic predictions across modalities using labels provided with the dataset, facilitating the vision and language embeddings to be mapped into a shared latent space:

$$\mathcal{L}_{cls} = \mathcal{L}_{cls_l}(y, \text{Classifier}(\hat{E}_l)) + \mathcal{L}_{cls_v}(y, \text{Classifier}(\hat{E}_v)) \quad (2)$$

KL-Divergence Minimization. To reduce distributional discrepancies between expanded vision and language embeddings, we applied:

$$\mathcal{L}_{kl-div} = D_{KL}(\hat{E}_v \parallel \hat{E}_l) \quad (3)$$

to ensure statistical similarity between modalities, complementing the embedding reconstruction task.

The overall training objective integrates reconstruction, classification, and distribution alignment losses into a unified optimization framework:

$$\mathcal{L} = \mathcal{L}_{rec} + \mathcal{L}_{cls} + \mathcal{L}_{kl-div} \quad (4)$$

By expanding embeddings rather than compressing them, CREED retains critical clinical semantics contained within radiology reports. The synergy of reconstruction, classification, and distribution alignment enables robust cross-modal knowledge transfer, empowering the vision model to outperform purely visual approaches that only learn from labels.

Implementation Details. The frozen Llama-3.1-8B [4] model serves as the teacher LLM, initialized with pretrained weights from the Meta Llama repository. For the student model, we adopted a Swin-Base Transformer (Swin-B) [11] pretrained on ImageNet-22k, using an input resolution of 224×224 . We replaced the final classification layer with a projector to align the output dimension with Llama’s hidden representation while retaining its hierarchical feature extraction capabilities. The encoder and decoder are implemented using multilayer perceptrons (MLPs), whereas the classifier is a single linear layer. We employed mean squared error (MSE) loss to enforce embedding reconstruction and binary cross-entropy (BCE) loss for multi-label classification for training objectives.

We utilized the official training split of the MIMIC-CXR dataset [8], consisting of 354,538 CXR images paired with their corresponding radiology reports, for pretraining. We extracted and processed the “Findings” and “Impression” sections of the reports, as they encapsulate radiologists’ diagnostic conclusions, providing rich textual inputs for the LLM. Additionally, we adopted the 14 structured labels officially released with the dataset (derived from the radiology reports) as ground-truth targets for cross-modal alignment.

3 Experiments and Results

3.1 CREED achieves SOTA performance in various diagnostic tasks

Experimental Setup. We conducted extensive experiments across diverse diagnostic tasks, including long-tailed, multi-class, multi-label, and binary classification to evaluate CREED. We benchmarked CREED against four comparative models: (1) a supervised baseline using an ImageNet-pretrained model [11], fine-tuned on the benchmark datasets; (2) the SOTA fully-supervised Ark-6 [13] model trained on 704K CXRs; (3) SimMIM-CXR [12]; and (4) RAD-DINO [21], both self-supervised models pretrained on 926K and 882K CXRs, respectively. RAD-DINO adopts a ViT-B backbone, whereas others use Swin-B. Downstream models initialized their encoders with pretrained weights and reinitialized task-specific classification heads according to target class number to ensure fair comparison. All layers were fine-tuned under identical experimental protocols following [12]. The experimental details for fine-tuning are available at [GitHub.com/JLiangLab/BenchmarkTransformers](https://github.com/JLiangLab/BenchmarkTransformers).

We employ four datasets to comprehensively evaluate CREED’s performance: ChestDR [25], which exhibits a long-tailed class distribution comprising 10 head classes and 9 tail classes; COVIDx [26], designed to distinguish among patients without pneumonia, with non-COVID-19 pneumonia, and with COVID-19 pneumonia cases; VinDrCXR [16], featuring six image-level global labels for multi-label classification; and ShenzhenCXR [6], focusing on screening tuberculosis (TB) from normal CXRs. Metrics used to assess model performance included AUC, Accuracy (ACC), F1 Score, Average Precision (AP), and Matthews Correlation Coefficient (MCC). Results are reported as means $\pm$ standard deviations over five independent runs on hold-out test sets. Statistical significance is assessed via independent two-sample t-tests ($p < 0.05$).

Results and Analysis. Table 1 presents the performance comparison between CREED and existing SOTA models across various thoracic disease diagnostic tasks. CREED achieves a mean AUC of 82.70% when fine-tuned at 224×224 resolution on ChestDR, significantly surpassing the ImageNet-pretrained baseline and the self-supervised SimMIM-CXR model, while performing comparably to the SOTA supervised Ark-6 and self-supervised RAD-DINO models. Notably, when fine-tuned at a higher resolution of 448×448 , CREED attains the highest AUC (83.38%), MCC (31.36%), AP (38.57%), and F1 score (34.14%), outperforming both Ark-6 and RAD-DINO. These results highlight CREED’s strong capacity for enhancing representation learning in long-tailed classification tasks. Additionally, CREED achieves the highest accuracy (97.75%) and MCC (96.89%) on COVIDx. CREED also demonstrates statistically comparable performance to Ark-6 on VinDrCXR and ShenzhenCXR, even when the latter had the advantage of incorporating training images and labels from these datasets during pretraining. These results collectively underscore CREED’s effectiveness for vision-based chest radiographic diagnosis across diverse classification tasks, further highlighting its capability to transfer knowledge from LLMs for robust cross-modal representation learning.

Table 1. CREED achieves SOTA performance across various thoracic disease diagnostic tasks, outperforming or matching existing SOTA fully- and self-supervised models, even those pretrained on larger multi-source datasets. This underscores the CREED’s effectiveness for achieving superior diagnostic accuracy, demonstrating the impact of cross-modal distillation from LLMs and radiology reports. Performance values are formatted as follows: bold for the best result, underlined for the second-best, and green shading to indicate methods with no statistically significant difference ($p \geq 0.05$) from the best result.

Method	Input resolution	AUC (%)	ChestDR		F1 (%)
			MCC (%)	AP (%)	
Baseline		77.39±0.48	25.19±0.52	28.41±0.62	29.34±0.39
SimMIM-CXR	224×224	78.03±0.95	25.25±0.92	28.06±1.01	28.99±0.58
Ark-6		82.58±0.12	31.01±0.21	37.03±0.36	33.82±0.26
CREED		82.70±0.10	30.53±0.58	37.05±0.85	33.32±0.70
RAD-DINO	518×518	82.73±0.57	31.10±1.16	37.24±1.90	33.89±1.11
CREED*	448×448	83.38±0.15	31.36±0.28	38.57±0.80	34.14±0.38

Method	Input resolution	COVIDx		VinDrCXR	ShenzhenCXR
		Accuracy (%)	MCC (%)	AUC (%)	AUC (%)
Baseline		96.10±0.83	92.28±0.33	90.35±0.31	93.35±0.77
SimMIM-CXR	224×224	96.17±0.56	93.88±2.02	91.71±1.04	95.76±1.79
Ark-6 [†]		97.25±0.40	95.34±0.57	95.07±0.16	98.98±0.15
CREED		97.40±0.34	95.30±0.80	94.06±0.18	98.28±0.36
RAD-DINO [‡]	518×518	97.05±0.27	95.58±0.81	94.74±0.35	98.66±0.08
CREED*	448×448	97.75±0.64	96.89±0.54	94.64±0.55	98.92±0.35

* Our experiments use the same pretrained model (initialized at 224×224 resolution), with input resolution increased to 448×448 during fine-tuning.

[†] Ark-6 incorporated training images and labels from the VinDrCXR and ShenzhenCXR datasets during its pretraining phase, which may confer a comparative advantage in downstream tasks.

[‡] RAD-DINO, continuously trained from DINOv2-Base pretrained on LVD-142M (a dataset vastly larger than ImageNet), leverages more CXR images and higher-resolution inputs for self-supervised pretraining, positioning it as a strong foundation model for comparison.

3.2 CREED boosts performance for radiology report generation

Experimental Setup. Automated radiology report generation has become a sophisticated yet highly valuable AI-driven task due to the advancement of LLMs. By generating coherent paragraphs that accurately capture the observations and findings from radiology images, this technology can significantly reduce radiologists’ workload and diagnostic errors, accelerating clinical workflows [2,27]. To demonstrate that CREED effectively distills detailed clinical knowledge from LLM embeddings into its vision model, we evaluate the model on radiology report generation using the MIMIC-CXR [8] dataset, following the framework and configuration of R2GenGPT [27]. R2GenGPT employs a visual alignment module, implemented as a linear layer, to map visual features from a visual model into the word embedding space of a frozen LLM, enabling the generation of coherent reports from radiographic images. We adopted the Shallow Alignment configuration to isolate the impact of knowledge distilled from the LLM during CREED’s pretraining, training only the linear alignment layer while keeping the vision model fixed. We also evaluated CREED under the Deep Alignment setup, where the entire model was fine-tuned. In both settings, the vision model was initialized with pretrained weights from CREED, ensuring a direct evaluation

Table 2. CREED improves report generation performance under both R2GenGPT [27] configurations. Notably, in the Shallow Alignment setting, with only 4.2M trainable parameters, CREED achieves performance comparable to the Deep Alignment baseline, which uses 90.9M trainable parameters.

Configuration	Param.	Model	Bleu-1	Bleu-2	Bleu-3	Bleu-4	ROUGE-L	CIDEr
Shallow Alignment	4.2M	R2GenGPT	0.364	0.235	0.161	0.116	0.276	0.135
		CREED	0.401	0.261	0.182	0.130	0.296	0.195
Deep Alignment	90.9M	R2GenGPT	0.406	0.268	0.186	0.135	0.298	0.262
		CREED	0.413	0.269	0.189	0.139	0.304	0.268

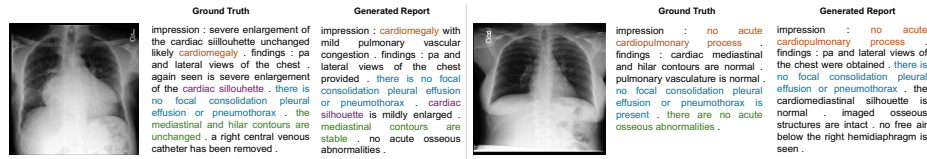


Fig. 3. Examples of radiology reports generated by R2GenGPT using CREED. Key medical findings are highlighted with colors for better illustration.

of its effectiveness in report generation. We employed several metrics to assess the quality of generated reports against radiologists’ written reports, including BLEU scores [20], ROUGE-L [9], and CIDEr [24], according to the established evaluation protocol [27].

Results and Analysis. Table 2 presents the comparison between the baseline R2GenGPT model and CREED across multiple text generation metrics, including BLEU-n, ROUGE-L, and CIDEr. CREED improves radiology report generation performance across all evaluated metrics under both configurations. Under the Shallow Alignment configuration, the BLEU-1 to BLEU-4 scores increase by 0.037, 0.026, 0.020, and 0.014, respectively, indicating enhanced unigram (word-level) precision and phrase-level coherence in the generated reports. ROUGE-L rises from 0.276 to 0.296, reflecting improved semantic continuity and structural alignment with human-written reports. Notably, CIDEr demonstrates a substantial improvement, increasing from 0.135 to 0.195, suggesting that CREED generates reports that are more informative, contextually relevant, and diverse. Deep Alignment can further improve the performance, especially for the CIDEr metric. Additionally, Fig. 3 illustrates two example reports generated using CREED, with key medical findings highlighted to demonstrate their alignment with expert-written reports. The results underscore the effectiveness of CREED’s cross-modal knowledge distillation, demonstrating its ability to bridge the gap between visual features and language embeddings, leading to improved performance in automated radiology report generation, enabling more accurate and coherent medical descriptions.

Table 3. Performance degradation caused by ablating key components.

Configuration	ChestDR (AUC%)	ShenzhenCXR (AUC%)
Full model (CREED)	82.70±0.10	98.28±0.36
No reconstruction loss	81.32±0.51 (↓1.38)	97.43±0.33 (↓0.85)
No classification loss	80.04±0.75 (↓2.66)	96.31±0.45 (↓1.97)
No KL-divergence loss	81.42±0.43 (↓1.28)	97.09±0.03 (↓1.19)

3.3 Ablation Study

We systematically ablate the reconstruction loss ($\mathcal{L}_{rec}$), classification loss ($\mathcal{L}_{cls}$), and KL-divergence loss ($\mathcal{L}_{kl-div}$) to quantify the contribution of each loss component in CREED. The results on ChestDR and ShenzhenCXR are summarized in Table 3. Removing any of these losses leads to a significant performance decrement, underscoring their complementary roles in effective cross-modal distillation. Notably, the reconstruction and classification losses work synergistically: reconstruction preserves fine-grained semantics, whereas classification ensures task-relevant alignment, reinforcing CREED’s robustness.

4 Conclusion and Future Work

We have introduced CREED, a novel cross-modal knowledge distillation framework that integrates radiology report semantics into vision-based CXR diagnosis models using embeddings from LLMs. CREED bridges the modality gap between vision and language by employing embedding reconstruction, diagnostic classification, and KL-divergence minimization. Our experiments show that CREED achieves SOTA performance across various diagnostic tasks and enhances report generation.

Future work will focus on incorporating additional datasets for large-scale pretraining, leveraging advanced medical LLMs as teacher models, and introducing explicit multimodal interaction mechanisms (*e.g.*, attention modules, fusion layers, and token-level correspondence) to further enhance model performance. Additionally, a benchmark needs to be established for evaluating the model’s ability to assess diagnostic uncertainty and disease severity, and to validate its effectiveness on real-world clinical cases.

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